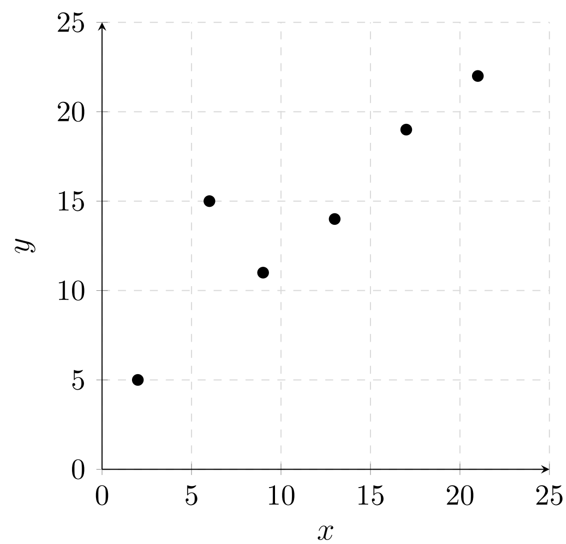
La Scuola d'Italia / Huson / IB Math: Linear Regression First & last name:
2 March 2026 11th Grade

2.7 Classwork: Linear Regression

1. The table shows six paired observations of variables x and y. [5 marks]

x	2	6	9	13	17	21
y	5	15	11	14	19	22

A scatter diagram of the data is shown below.



(a) On the scatter diagram above, draw a line of best fit. [1]

(b) Using your line, write down the approximate equation in the form $y = ax + b$. [1]

(c) Use your GDC to find: [2]

(i) the equation of the regression line of y on x, giving a and b correct to three significant figures;

(ii) the value of Pearson's product-moment correlation coefficient, r.

(d) Use the guidance below to describe the correlation between x and y. [1]

$0 \le |r| < 0.4$: weak, $0.4 \le |r| < 0.8$: moderate, $0.8 \le |r| \le 1$: strong.

2. Eight Italian cities were surveyed to investigate the relationship between latitude and average annual salary. The data are shown in the table below. [7 marks]

City	Latitude, L (°N)	Average salary, S (€/year)
Milano	45.5	37,189
Torino	45.1	26,840
Bologna	44.5	29,300
Genova	44.4	27,000
Firenze	43.8	28,646
Roma	41.9	28,000
Napoli	40.9	24,000
Palermo	38.1	22,000

(a) Find the value of Pearson's product-moment correlation coefficient, r. [1]

(b) Use the guidance below to describe the correlation between latitude and average annual salary. [1]

$$0 \leq |r| < 0.4\text{: weak,} \quad 0.4 \leq |r| < 0.8\text{: moderate,} \quad 0.8 \leq |r| \leq 1\text{: strong.}$$

(c) Write down the equation of the regression line of S on L, in the form $S = aL + b$, where a and b are given to three significant figures. [2]

(d) Bari is an Italian city at latitude 41.1°N. Use your regression line to estimate the average annual salary in Bari. [2]

(e) Interpret the gradient of your regression line in the context of this problem. [1]